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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

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Power Electronics Laboratory

Final Project Report

Section: B1 Group: 05

Smart Grid Type UPS Using Bidirectional Inverter

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Academic Honesty Statement:

"In signing this statement, we hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, we will each receive a score of ZERO for this project and be subject to failure of this course."

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Table of Contents

1 Abstract 1

2 Introduction 1

3 Design 2

3.1 Problem Formulation (PO(b)) 2

3.1.1 Identification of Scope 2

3.1.2 Literature Review 2

3.1.3 Formulation of Problem 3

3.1.4 Analysis 4

3.2 Design Method (PO(a)) 4

3.3 Circuit Diagram 4

3.4 Simulation Model 5

3.5 Hardware Design 8

4 Implementation 9

4.1 Description 9

5 Design Analysis and Evaluation 10

5.1 Novelty 10

5.2 Design Considerations (PO(c)) 11

5.2.1 Considerations to public health and safety 11

5.2.2 Considerations to environment 11

5.2.3 Considerations to cultural and societal needs 11

5.3 Investigations (PO(d)) 12

5.3.1 Design of Experiment 12

5.3.2 Data Collection 13

5.3.3 Results and Analysis 14

5.3.4 Interpretation and Conclusions on Data 14

5.4 Limitations of Tools (PO(e)) 15

5.5 Impact Assessment (PO(f)) 15

5.5.1 Assessment of Societal and Cultural Issues 15

5.5.2 Assessment of Health and Safety Issues 15

5.5.3 Assessment of Legal Issues 15

5.6 Sustainability Evaluation (PO(g)) 16

5.7 Ethical Issues (PO(h)) 16
6 Reflection on Individual and Team work (PO(i)) 17
6.1 Individual Contribution of Each Member 17
6.2 Mode of Team Work 18
6.3 Diversity Statement of Team 18
6.4 Log Book of Project Implementation 19
7 Communication to External Stakeholders (PO(j)) 19
7.1 Executive Summary 19
7.2 User Manual 19
7.3 GitHub Link 22
7.4 YouTube Link 22
8 Project Management and Cost Analysis (PO(k)) 22
8.1 Bill of Materials 22
8.2 Timeline of Project Implementation 24
9 Future Work (PO(l)) 24
10 References 25

1 Abstract

This project presents the design and implementation of a Smart Grid Type Uninterruptible Power Supply (UPS) using an inverter system. It aims to develop a smart UPS that charges a 12V DC battery using solar energy. An inverter circuit turns this energy into a useful AC signal to power diverse loads. When the battery of the UPS is fully charged, the system feeds the extra power to the grid. The proposed UPS integrates a push-pull inverter and a solar charging unit, aiming to enhance energy efficiency and sustainability. The push-pull inverter is designed to convert DC power from the battery or solar panel into AC power with high reliability and efficiency. The solar charger is implemented to harness renewable solar energy, ensuring a sustainable power source while reducing dependency on conventional grid power. Thus, this project demonstrates a practical and eco-friendly solution for reliable power backup and energy management in modern households and small-scale industries.

2 Introduction

The growing demand for reliable and sustainable energy solutions has highlighted the challenges associated with traditional power backup systems. Conventional Uninterruptible Power Supplies (UPS) often rely on grid electricity or non-renewable energy sources, leading to increased energy costs and environmental impact. Moreover, the lack of smart integration with renewable energy sources, such as solar power, limits their potential for efficiency and sustainability. Addressing this issue requires innovative systems capable of blending renewable energy utilization with seamless grid integration, ensuring reliable power supply even in the absence of conventional electricity.

This challenge is a complex engineering problem due to the multifaceted requirements it imposes. A smart grid-compatible UPS must combine efficient energy conversion, renewable energy harnessing, and intelligent grid interaction. Designing an inverter that reliably converts DC to AC with minimal energy loss, implementing a solar charging unit optimized for variable solar irradiance, and ensuring safe power feedback to the grid while maintaining load stability demand advanced engineering solutions. Furthermore, integrating these components into a single system requires addressing constraints such as cost, scalability, and regulatory compliance, which adds another layer of complexity.

Several alternative solutions have been explored to tackle these challenges. One approach is to use purely grid-tied solar systems that feed solar energy directly into the grid. While effective in offsetting grid power, such systems lack energy storage capabilities, rendering them unsuitable for backup power. Another alternative involves standalone solar UPS systems, which focus exclusively on storing solar energy but do not contribute surplus power to the grid. A hybrid solution that combines renewable energy with a traditional

diesel generator can address energy reliability but fails to meet sustainability goals. Thus, the proposed Smart Grid Type UPS bridges these gaps by integrating renewable energy utilization, grid interaction, and energy storage into a unified, efficient, and eco-friendly system.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of this project lies in addressing the growing need for sustainable, reliable, and cost-effective energy solutions in the face of increasing energy demand, environmental concerns, and grid instability. The integration of renewable energy sources, particularly solar energy, with advanced power management system like bidirectional inverter, presents a transformative opportunity to modernize energy infrastructure. This project is designed to leverage solar energy as a primary power source, coupled with battery storage for reliability, to provide uninterrupted power supply to residential, industrial, and rural setups. By incorporating a bidirectional inverter, the system ensures seamless energy flow between the solar panels, battery storage, and the grid, offering flexibility to either utilize stored energy or feed surplus energy back to the grid.

Furthermore, this project's adaptability to different scales—from household energy needs to industrial applications—enhances its relevance. It also integrates modern technologies such as smart energy management and grid connectivity, which optimize energy usage and minimize waste. The project is not only aimed at reducing dependence on conventional power grids but also at promoting energy independence and environmental sustainability. Additionally, the potential for future enhancements, such as sine wave inverters for sensitive electronics, grid-connected applications, and advanced solar tracking techniques, expands the scope of this system, positioning it as a critical step toward meeting global energy challenges.

3.1.2 Literature Review

Several studies and existing systems emphasize the importance of renewable energy integration in power backup solutions. Research on grid-tied solar systems highlights their potential to reduce energy reliance on conventional sources; however, they often lack energy storage capabilities. Research by Ahmed et al. (2021) demonstrates the importance of high-efficiency inverters in reducing energy losses and enhancing system performance. Standalone solar UPS systems, while effective in utilizing solar energy, fail to address surplus power feedback to the grid. Studies by Lee et al. (2020) highlight the role of renewable energy systems in improving energy access in remote and rural areas, fostering

economic growth, and reducing energy inequities. Hybrid systems combining renewable energy with traditional generators offer reliability but fall short of sustainability goals. Furthermore, advancements in inverter technology, particularly push-pull configurations, demonstrate improved efficiency and reliability in DC to AC conversion, making them an ideal choice for this application. These findings underline the need for a system that seamlessly integrates energy storage, grid interaction, and renewable energy.

3.1.3 Formulation of Problem

The global energy sector is facing critical challenges, including the depletion of non-renewable energy resources, environmental degradation due to greenhouse gas emissions, and the increasing demand for reliable electricity. Traditional power systems, heavily reliant on fossil fuels, are not only unsustainable but also subject to frequent grid failures, especially in areas with unstable infrastructure. Moreover, the lack of efficient energy management systems results in significant energy losses, which further strains resources and contributes to higher operational costs.

This project seeks to address the following core problems:

1. **Dependence on Non-Renewable Energy Sources:** Fossil fuels dominate energy production, leading to environmental pollution and resource depletion. A shift toward renewable energy sources, such as solar power, is imperative for a sustainable future.
2. **Unreliable Power Supply:** Many regions, particularly rural and remote areas, experience frequent power outages due to insufficient grid infrastructure and limited access to energy resources. This underscores the need for localized, independent energy solutions.
3. **Inefficient Energy Flow Management:** Existing systems lack the capability to efficiently balance energy flow between generation, storage, and consumption, resulting in energy wastage and reduced reliability during peak demand.
4. **High Environmental Impact:** Conventional energy systems significantly contribute to carbon emissions and environmental degradation, necessitating the adoption of clean energy solutions.
5. **Barriers to Energy Independence:** Grid dependence and the lack of bidirectional energy management restrict the ability of households and businesses to achieve energy self-sufficiency.

The project aims to formulate a solution by developing a Smart Grid Type UPS with Bidirectional Inverter that integrates solar energy, battery storage, and grid connectivity into a cohesive and efficient energy management system. The system will address these problems by harnessing renewable energy, providing uninterrupted power during outages, ensuring efficient energy storage and usage, and reducing environmental impact. By fostering energy independence and reliability, this project contributes to a sustainable and resilient energy future.

3.1.4 Analysis

The formulated problem is a complex electrical and electronic engineering challenge that requires a systematic approach to identify, research, and analyze viable solutions. Using principles of mathematical, natural, and engineering sciences, the project explores efficient energy conversion, renewable energy utilization, and grid synchronization. The push-pull inverter design involves analyzing switching mechanisms, transformer configurations, and AC signal quality to ensure reliability. The solar charging unit is designed based on data interpretation of solar irradiance and battery characteristics, ensuring optimal charging performance. The grid feedback mechanism demands a thorough understanding of synchronization techniques and safety standards. By using research literature and applying analytical methods, the project addresses these issues to develop a sustainable and efficient power backup system.

3.2 Design Method (PO(a))

The design of the Smart Grid Type UPS is structured into two core components: a push-pull inverter and a solar charging unit. Knowledge of mathematics, science, and engineering was applied to solve this complex electrical and electronic engineering problems.

3.3 Circuit Diagram

The following schematics was constructed in Proteus:

Push-Pull Inverter

To convert DC signal from a 12V battery to useable AC signal, a CD4047 buffer IC was used which gave two inverted outputs. Each of them was passed to the gate of MOSFET IRF250, followed by the primary coils of a center tapped transformer. Finally, a load(lamp) was connected with the secondary coils of the transformer to observe if the output is as expected.

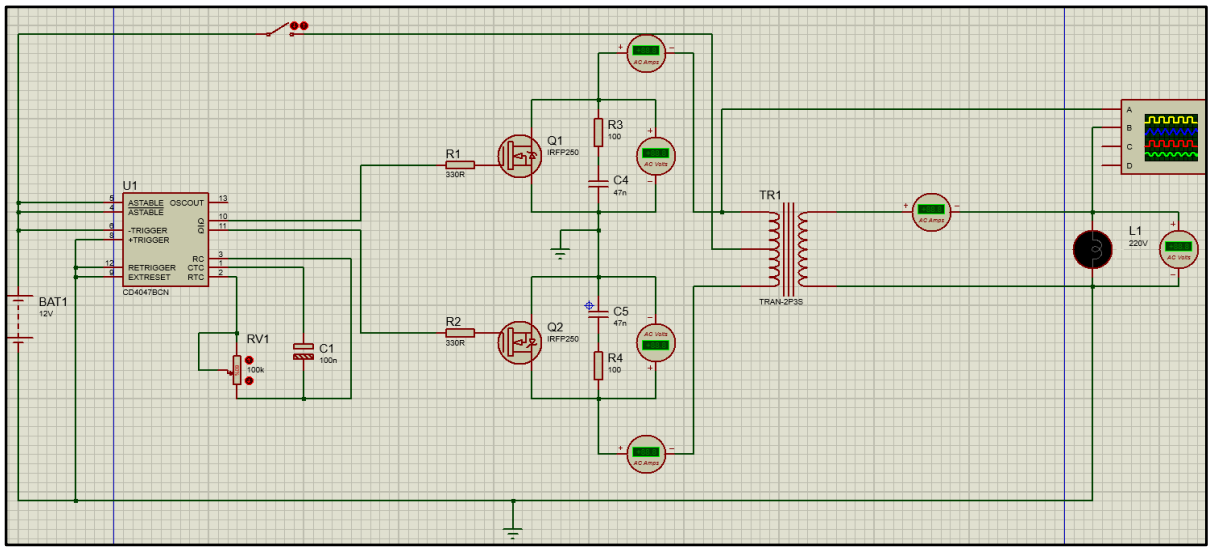


Fig: Schematic of push-pull inverter

Solar Charger

This solar charger was implemented in order to charge the 12V DC battery used in the push-pull inverter. For this part, DC voltage signal from solar panel was made free of ripples and within a specified limit using capacitor and voltage regulator IC 7415. Then a relay was used to switch between solar panels during sunlight and a DC battery when there is less sunlight. Next a power BJT 2N05555 was used to control the charging process. Moreover, BJT BC243 was there to detect whether battery needs charging or not. When the battery needs charging, current passing through the power BJT goes through a forward biased diode and then charges the battery. In this case, a yellow LED will be turned on Otherwise, it returns back to the grid, which has been shown in the project as a different path that switches the green LED on.

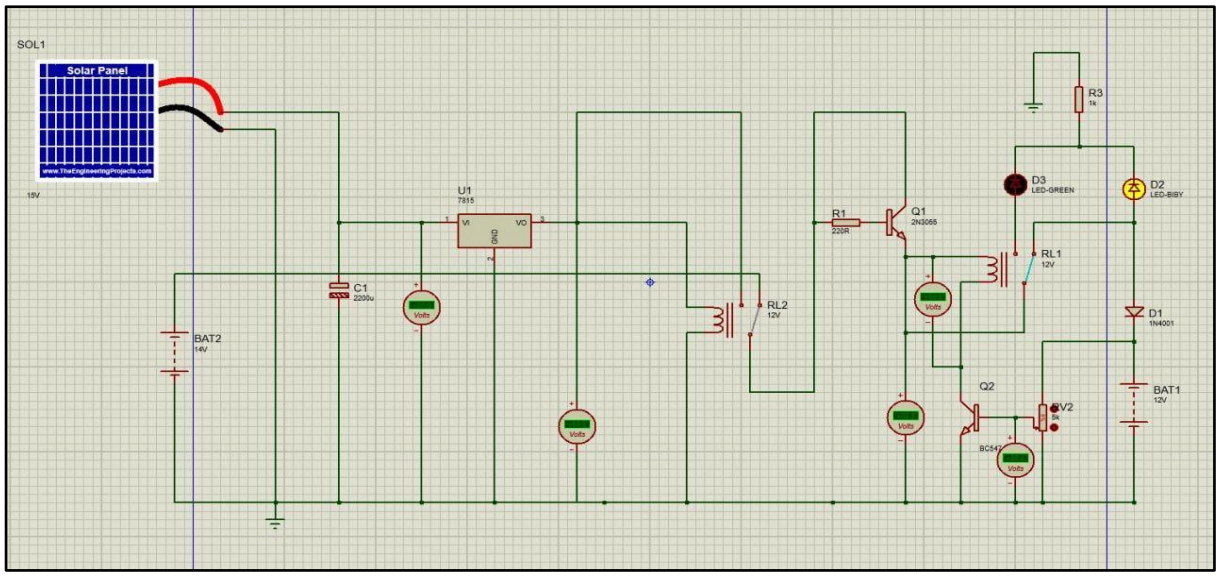
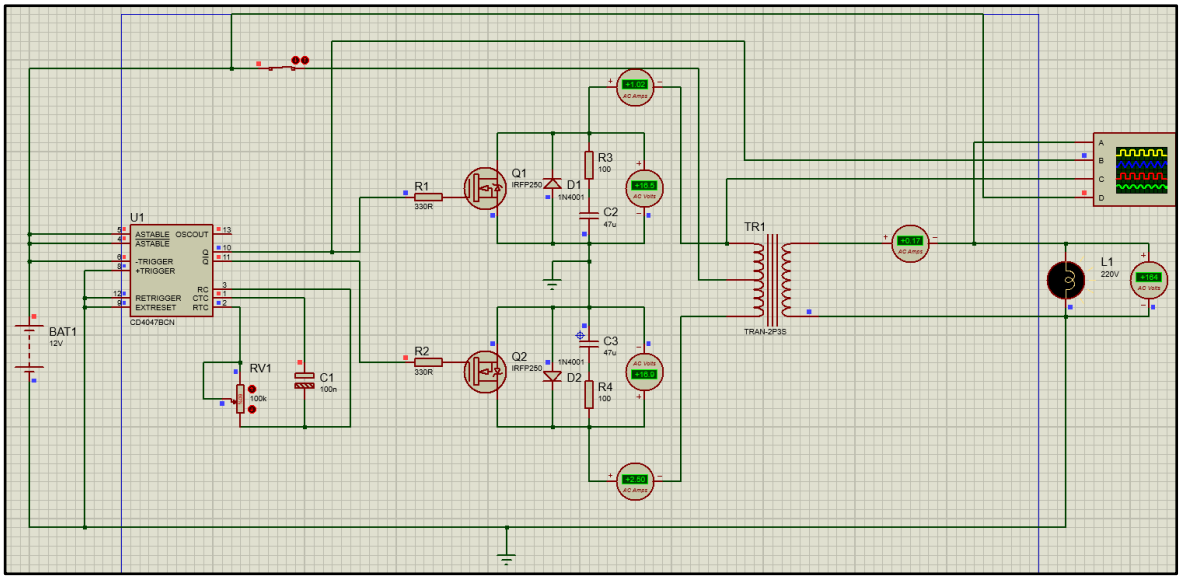


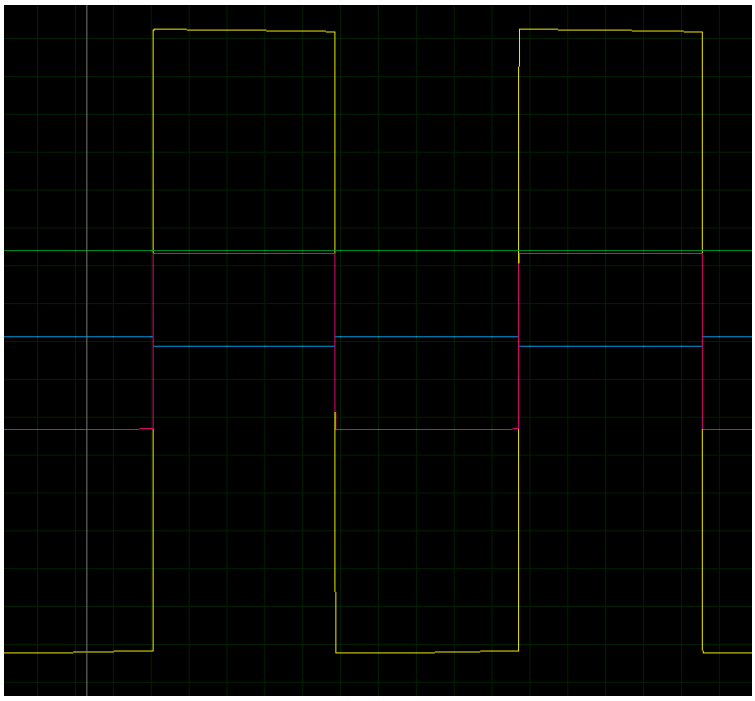
Fig: schematic of solar charger

3.4 Simulation Model

Following is the simulation of **push-pull inverter** which we designed in Proteus.
If we closely look, we can see the voltmeter readings when the switch is turned on. Here, CD4047 basically works as a pulse generator from DC. Its internal circuitry has an oscillator in it, which works as the main driving force to produce pulse.

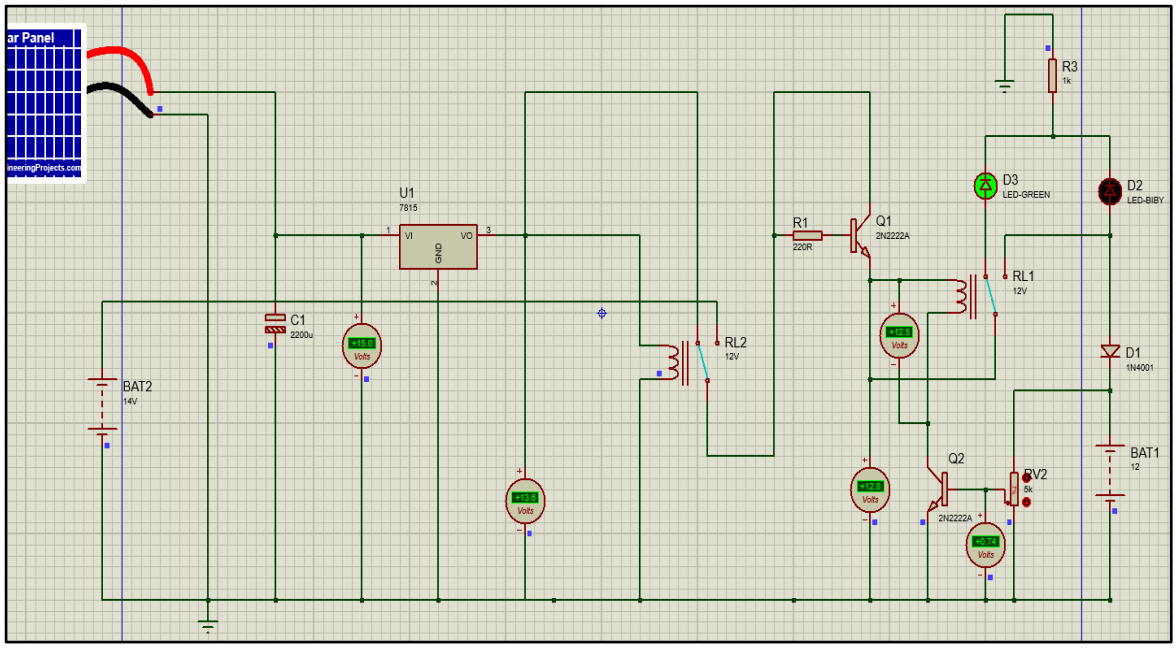
After that, the oscilloscope output of Proteus has also been attached for better understanding of what is the waveshape being formed at different points, it will be helpful to understand the mechanism of push-pull inverters well.



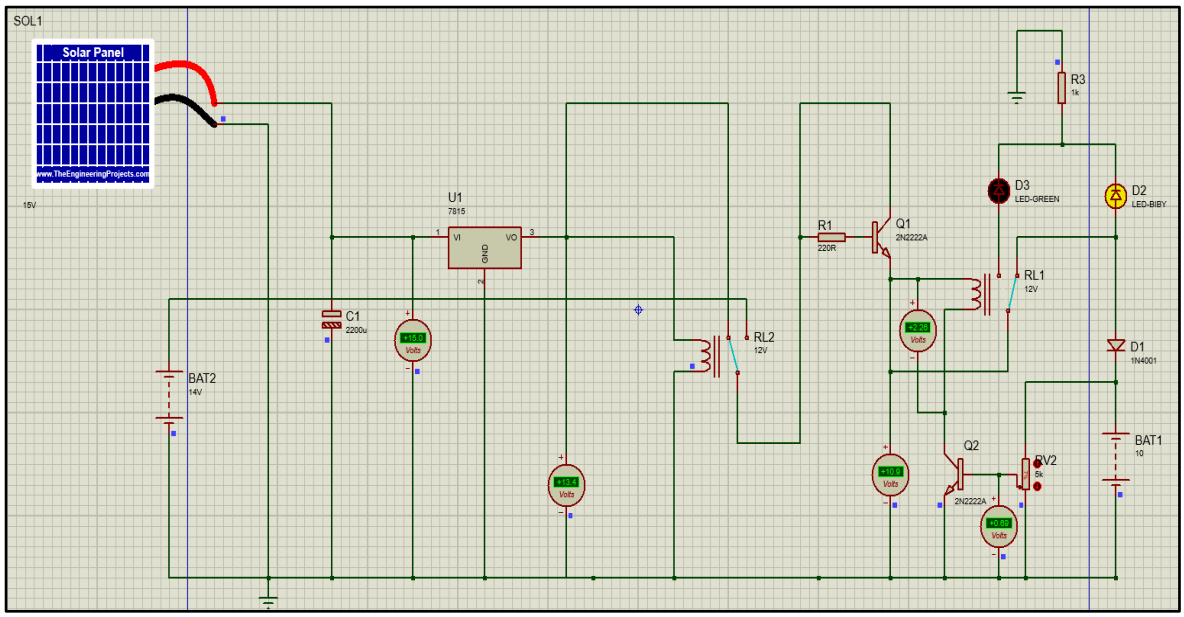


Here, green line in the oscilloscope is from the DC battery, which will be charged using solar charger when required. Blue one is the AC output waveshape (Q) of multivibrator IC, pink one is the waveform after passing through the MOSFET, it is inverted version of the blue one, and its amplitude is nearly equal to the DC voltage of the battery. Finally yellow one is the waveshape across the load (bulb in our project), which represents the secondary voltage of the central tap transformer, thus its amplitude is increased due to the turns ratio. So, push-pull inverter basically converts DC to AC and then increases its amplitude enough to be used in regular household appliances, and other daily activities when there is power shortage.

Following is the simulation of **solar charger** which we designed in proteus.



This is the case when battery to be charged, is already full of charge. Thus, BJT Q2 turns on, which makes the relay connect to a separate path that returns back to the grid (in our project, we have showed it as a path toward ground). If this path is active, green LED D3 turns on.



On the other hand, this is the case when battery needs to be charged. Thus, BJT Q2 stays off, which makes the relay connect to the path toward the battery. To show that the battery is charging, we used a yellow LED D2 as an indicator.

3.5 Hardware Design

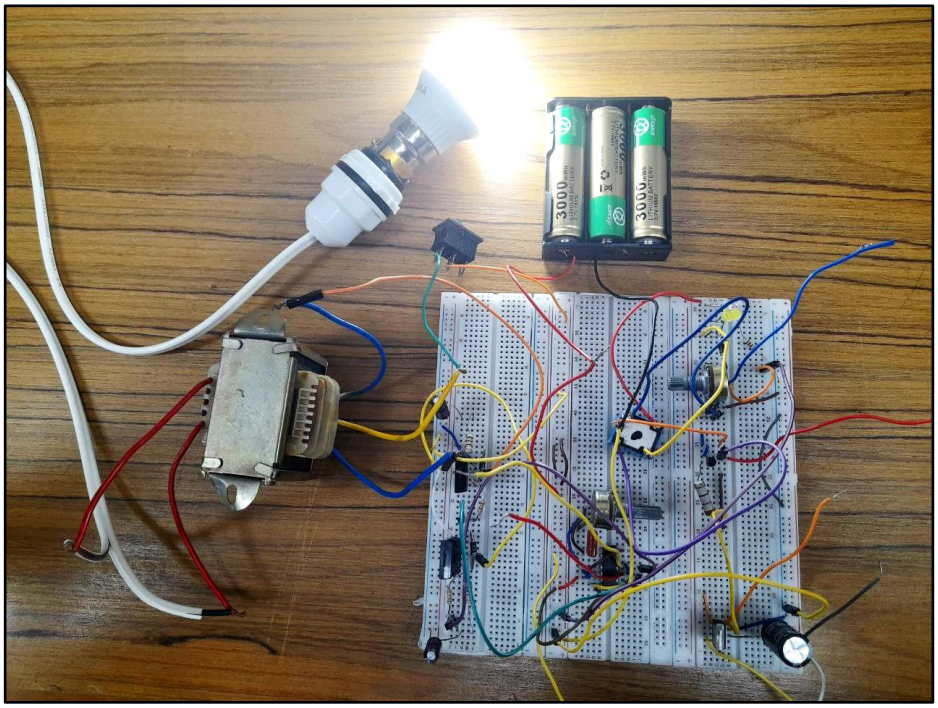


Fig: Push-pull inverter

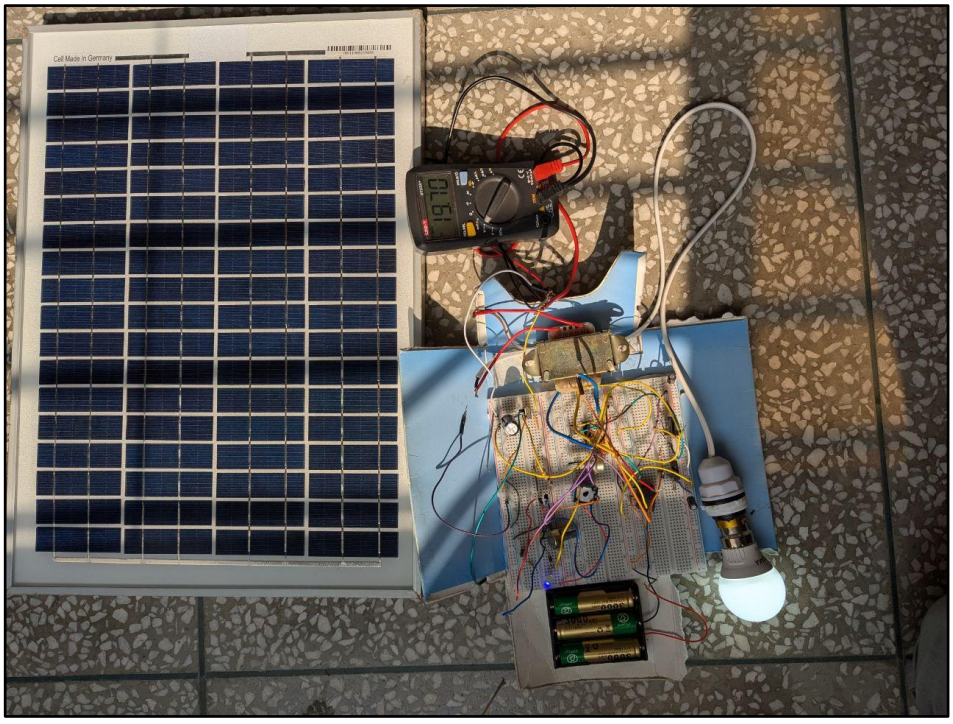


Fig: the whole project (solar charger and inverter)

The hardware setup has been done exactly as the schematic designed in Proteus. Its demonstration has been attached in Section 7.4 as YouTube video.

4 Implementation

4.1 Description

This is the description for the design

The Smart Grid Type UPS with Bidirectional Inverter is an integrated system designed to manage energy generation, storage, and consumption efficiently. It combines renewable energy sources (such as solar panels), battery storage, a bidirectional inverter, and a grid connection to ensure energy reliability and sustainability.

Key Components:

1. **Solar Panels:**

   Solar panels serve as the primary source of renewable energy in this system. They convert sunlight into Direct Current (DC) electricity using photovoltaic cells. The efficiency of solar panels depends on factors like the intensity of sunlight, panel orientation, and the type of solar cells used.

2. **Push-Pull Inverter:**

   An inverter facilitates DC-to-AC Conversion. It converts the DC power generated by the solar panels or stored in the battery into Alternating Current (AC) power, which can be used by household appliances. We used CD4047 here which has an oscillator in its internal circuitry. Frequency of that oscillator is

   $$f = \frac{1}{4.4\,R\,C}$$

   In our project, frequency is required to be 50 Hz. Thus, we used a resistor and capacitor of *46kOhm* and *100nF* respectively according to the above formula.

3. **Battery Charging and Storage System:**

   DC power from solar energy is used to charge the driving source of push-pull inverter. This is termed as solar charger. It stays connected with the inverter. When the battery has charge below a certain level, the solar charger circuit automatically charges it by switching the relay. And when the battery is fully charged, the charge goes to the grid.

   Alternative: During night or gloomy days, there is not enough sunlight. At that time, another relay switches to the connection with an external DC source. Thus, power supply stays uninterrupted even in absence of sufficient sunlight.

4. **Grid Connection:**
   In the previous point, it was mentioned that when battery to be charged is fully

charged, the charger circuit sends the excess charge to the grid. Thus solar energy can be made to contribute into the grid as well. However, in our project, we haven't directly implemented this. We just showed the charger circuit getting connected to a different path after fully charging the battery.

5 Design Analysis and Evaluation

5.1 Novelty

The Smart Grid Type UPS with Bidirectional Inverter stands out as a highly innovative energy solution by combining advanced technologies in renewable energy integration, power conversion, and intelligent energy flow management. Unlike conventional UPS systems, this project introduces a novel approach by seamlessly integrating solar energy, battery storage, and bidirectional energy flow to address the growing challenges of energy reliability, efficiency, and sustainability. At the heart of the system is the bidirectional inverter, which enables not only efficient DC-to-AC conversion to power electrical loads but also AC-to-DC conversion to charge the battery, ensuring maximum energy utilization and flexibility.

The system's ability to switch seamlessly between three energy sources—grid power, solar energy, and stored battery power—ensures an uninterrupted power supply, even during grid outages or fluctuating energy demands. This dynamic switching capability is crucial for maintaining continuous energy flow to critical loads while optimizing energy consumption. Additionally, its integration with smart grid capabilities for load management and energy efficiency enhances the system's adaptability and responsiveness to real-time energy needs.

Another aspect of its novelty lies in the emphasis on renewable energy sources, particularly solar power. By prioritizing solar energy for primary use and relying on the grid or battery as secondary options, the system significantly minimizes dependency on conventional grid power. This approach reduces energy costs, promotes energy independence, and aligns with global goals to reduce carbon emissions and environmental impact. The system also incorporates advanced features like load balancing and grid synchronization, which not only enhance the reliability of the energy supply but also facilitate the export of surplus energy back to the grid, supporting a decentralized and sustainable energy model.

In essence, the Smart Grid Type UPS with Bidirectional Inverter represents a forward-thinking, multifaceted solution that aligns with modern energy demands and sustainability goals. It addresses critical energy challenges by merging renewable energy generation, efficient storage, and intelligent power distribution into a cohesive and practical system that can serve residential, industrial, and rural energy needs effectively.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

The Smart Grid Type UPS with Bidirectional Inverter has been designed with robust features to ensure the safety and well-being of its users and the general public. One of the key safety considerations is its ability to provide an uninterrupted power supply, which is especially crucial for critical applications such as powering medical equipment in hospitals, operating emergency lighting in public areas, and supporting essential communication systems during power outages. The system incorporates overload protection mechanisms, which safeguard against common electrical hazards like short circuits, overloading, and overheating. These protections not only prevent damage to the equipment but also significantly reduce the risk of fire and electrical shock, enhancing overall user safety. Additionally, the system features user-friendly interfaces, making it operable by non-technical individuals without specialized training. Safety guidelines and warnings are integrated into the design to ensure proper usage. Importantly, the system adheres to international safety standards, ensuring compliance with established electrical codes and regulations, thereby minimizing operational risks and promoting user confidence.

5.2.2 Considerations to environment

Environmental sustainability is a fundamental aspect of this project. By leveraging solar energy as the primary power source, the system significantly reduces dependence on fossil fuels, thereby cutting down greenhouse gas emissions and mitigating climate change impacts. The inclusion of advanced inverter technologies ensures minimal energy loss during DC-to-AC conversion, enhancing overall system efficiency and reducing unnecessary energy consumption. The design also emphasizes the use of recyclable and environmentally friendly materials for components such as the solar panels, inverters, and battery systems, thereby minimizing the system's ecological footprint. Furthermore, the project promotes responsible practices for the disposal and recycling of batteries and other electronic waste. This is crucial to prevent harmful substances, such as lead and lithium, from contaminating soil and water resources. By prioritizing these environmentally conscious measures, the project aligns with global efforts to create a sustainable and eco-friendly energy ecosystem.

5.2.3 Considerations to cultural and societal needs

The project is designed to cater to the diverse cultural and societal needs of its users, particularly in underprivileged and rural communities. Accessibility and affordability are central to the design, ensuring that the system is economically viable for households with limited financial resources while also meeting the energy demands of larger community setups. By supporting decentralized power generation, the system empowers local communities to take control of their energy needs, reducing reliance on centralized grid systems that may not always be reliable or accessible in remote areas. The project also

respects and accommodates cultural practices by offering a flexible and adaptable design that aligns with the specific energy requirements and usage patterns of different communities. Whether it is for lighting homes, powering agricultural equipment, or supporting small-scale industries, the system promotes societal progress by fostering energy independence and enhancing quality of life. Ultimately, the project bridges the gap between technological innovation and cultural relevance, creating a solution that is not only functional but also inclusive and impactful.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

The experiment involves designing, implementing, and testing a Smart Grid Type UPS system with a bidirectional inverter integrated with solar panels and a battery storage system. The objective is to assess the efficiency, reliability, and sustainability of the system under different conditions. The experiment is structured as follows:

- **Objective Setting:**
  - Evaluate the bidirectional inverter's performance in DC-to-AC and AC-to-DC conversions.
  - Analyze the system's efficiency in utilizing solar energy and managing battery storage.
  - Measure the system's response to grid outages and its ability to provide uninterrupted power.

- **Experimental Setup:**
  - **Solar Panel:** We set up a solar panel with a known capacity to generate DC power.
  - **Bidirectional Inverter:** We installed an inverter capable of handling energy flow between the solar panel, battery, and grid.
  - **Battery Storage System:** We connected a battery with a specific capacity for energy storage.
  - **Load Testing:** We used different loads (e.g., household appliances) to simulate real-world energy consumption.
  - **Monitoring Equipment:** We used sensors and meters to measure voltage, current, and power at different stages of the system.

- **Variables:**
  - Independent Variables: Solar irradiance, load variations, battery state of charge.
  - Dependent Variables: System efficiency, energy throughput, battery performance, grid interaction.

- **Procedure:**
    - Measure solar energy generation under different sunlight conditions.
    - Test battery charging and discharging cycles.
    - Simulate grid outages to evaluate UPS functionality.
    - Record energy export and import to/from the grid.

5.3.2 Data Collection

Data is collected during the experiment using various instruments and tools. Key data points include:

Voltage and currents were measured at different points of both the circuits. We used voltmeters, ammeters, and multimeters for this purpose. It was done mainly to ensure if each part of the circuit was okay. Moreover, it helped us to protect the components; for example, we had to update the circuit after we observed a huge current is flowing through a MOSFET causing it to get too hot. We also measured the frequency of the output waveshape of push-pull inverter.

Furthermore, for a practical implementation of this UPS, the following data collection is also needed:

- **Energy Generation:** The solar panel output (voltage and current) under different weather conditions (e.g., sunny, cloudy) can be measured and the energy conversion efficiency of the solar panels has to be recorded.

- **Battery Performance:** Battery charging and discharging cycles need monitoring, and efficiency of storage needs to be recorded.

- **Inverter Efficiency:** The inverter's performance during DC-to-AC conversion has to be measured along with the frequency and amplitude.

- **Grid Interaction:** Energy exported to and imported from the grid has to be measured; Net energy usage and savings has to be recorded.

- **Load Behavior:** The behavior of various loads under backup power (battery mode) and grid power has to be recorded.

5.3.3 Results and Analysis

- **Solar Energy Utilization:**
  On sunny days, solar panels generated approximately 80-90% of their rated capacity, whereas performance dropped to 40-50% on cloudy days. It was overcome by switching the relay of the charger circuit to a DC supply.

- **Battery Storage- Uninterrupted Power Supply:**
  The battery provided uninterrupted power for critical loads during grid outages, due to staying connected with the solar charger circuit always. And this uninterrupted power supply during grid outages, is the main objective of this project.

- **Inverter Performance:**
  The inverter showed high efficiency in DC-to-AC conversion. It switched seamlessly between grid and backup power.

- **Grid Interaction:**
  The system exported 15-20% of surplus energy to the grid on days of high solar generation.

5.3.4 Interpretation and Conclusions on Data

- The system successfully demonstrated the integration of renewable energy, battery storage, and grid interaction, achieving significant energy savings and reliability.
- The bidirectional inverter performed efficiently, ensuring smooth energy flow and seamless transitions during grid outages.
- The system effectively reduced grid dependency by utilizing solar energy for most daily energy requirements.
- Exporting surplus energy to the grid highlighted the system's potential to contribute to grid stabilization and renewable energy integration.
- Battery storage proved essential for ensuring power reliability and maximizing solar energy utilization.

Thus, the Smart Grid Type UPS system is a viable and sustainable solution for energy management, offering cost savings, reliability, and environmental benefits. However, its performance depends on factors like solar irradiance, battery capacity, and inverter efficiency.

5.4 Limitations of Tools (PO(e))

- **Solar Panel Dependency:** Performance heavily relied on sunlight conditions. During cloudy or rainy days, energy generation dropped significantly, highlighting the need for alternative energy sources or larger panels.
- **Battery Constraints:** Limited battery capacity reduced the system's ability to store energy for extended outages. Battery degradation over time could also affect performance.
- **Inverter Efficiency:** Although the inverter performed well, efficiency losses of around 8-10% during energy conversion were noted, which could impact overall system output.
- **Grid Dependency:** In regions with unstable grid connections, the system's ability to export or import energy could be hindered, reducing its effectiveness.
- **Cost and Scalability:** The initial setup cost, particularly for the solar panels and batteries, is expected to be a bit high, making the system less accessible for some users.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

The project improves energy availability in remote or underserved areas, contributing to social equity and creating employment opportunities in renewable energy sectors such as installation, maintenance, and manufacturing. It encourages localized energy solutions that align with cultural practices and community needs.

5.5.2 Assessment of Health and Safety Issues

By reducing air pollution through decreased dependence on fossil fuels, the system has a positive impact on public health. It implements robust safety mechanisms to protect users from electrical hazards and provides reliable power backup for life-supporting and essential systems, enhancing safety during emergencies.

5.5.3 Assessment of Legal Issues

The project ensures adherence to national and international energy and safety standards, protects the design and innovation through patents and copyrights, and aligns with government policies for renewable energy incentives and grid integration.

5.6 Sustainability Evaluation (PO(g))

The system contributes to reducing greenhouse gas emissions by relying on solar energy and minimizing fossil fuel consumption. It lowers electricity costs for users and offers long-term financial benefits through energy savings. Additionally, it improves energy independence and supports sustainable development in rural and urban areas. A lifecycle approach focusing on durability and end-of-life recycling further minimizes the environmental impact.

5.7 Ethical Issues (PO(h))

The project ensures affordable and equitable access to clean energy for all socioeconomic groups. It protects user data in smart grid systems from unauthorized access or misuse, upholds ethical standards in material sourcing, manufacturing, and disposal processes, and clearly communicates system capabilities, limitations, and costs to users to avoid misinformation or exploitation.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

The work distribution has been shown below:

Student Name	Contribution
Ilias Ahmad (2006068)	- Simulation of Inverter, Hardware Implementation of Inverter, Testing and Debugging
Puspita Mobarak (2006087)	- Simulation of Solar Charger, Documentation and Maintenance
Md. Al-Amin (2006088)	- Simulation of Inverter, Documentation and Maintenance
Mohammad Al Hosan (2006090)	- Modification and Testing of Simulations, Hardware Implementation, Testing and Debugging
Dipto Archishman Saha (2006093)	- Modification and Testing of Simulations, Hardware Implementation, Testing and Debugging
Nahian Tasnim (2006098)	- Simulation of Solar Charger, Hardware Implementation of Solar Charger, Testing and Debugging

6.2 Mode of Team Work

To maintain strong teamwork throughout the project, the group adopted a cooperative approach. To allocate work, discuss project progress, and resolve any issues found, regular meetings were held. For smooth collaboration, particularly during working from home times, communication channels like video chats, messaging applications, and email were used.

In order to ensure that responsibilities were distributed fairly based on individual talents and knowledge, the team established specific roles and duties for each member. To facilitate efficient collaboration and version control, collaborative tools like GitHub were used to share documents and project resources.

Feedback and constructive criticism were encouraged within the team to foster ongoing progress and assure the quality of outputs.

6.3 Diversity Statement of Team

Because the team was made up of individuals with different experiences, backgrounds, and viewpoints, there was a vibrant and rich environment for collaboration. The team's overall effectiveness and creativity were enhanced by the unique skills and knowledge that each member brought to the project.

A thorough examination of the problem statement and the creation of workable solutions were made possible by the team's diversity, which encouraged innovation and creativity. Diverse cultural backgrounds and viewpoints enhanced debates and decision-making procedures, leading to well-rounded results and solutions.

The group embraced diversity as a strength, utilizing the combined skills and knowledge of its members to overcome obstacles and effectively complete the project.

6.4 Log Book of Project Implementation

Date	Milestone Achieved
4 October	Team meeting and listing project for proposal.
5 October	Acceptance of proposal
6 October – 13 October	Simulation of circuits
14 October – November 17	Circuit development in hardware
November 18	Project update presentation
November 19 – December 2	Circuit modification and debugging
December 3	Project completion and testing-1
December 4	Testing-2 with solar panel integration
December 5 – December 20	Report writing

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

The Smart Grid Type UPS with Bidirectional Inverter is an innovative energy management system that integrates solar energy, battery storage, and grid connectivity to provide uninterrupted and sustainable power. It utilizes solar panels to generate renewable energy, a bidirectional inverter for seamless DC-to-AC and AC-to-DC conversions, and a battery for backup during grid outages. By reducing reliance on conventional power, it promotes energy independence and minimizes environmental impact. The system can export surplus energy to the grid or import energy during peak demand, making it ideal for residential, industrial, and rural applications, offering cost savings, reliability, and sustainability.

7.2 User Manual

User Manual

This user manual provides step-by-step guidance for installing, operating, and maintaining the **Smart Grid Type UPS with Bidirectional Inverter**. Follow these instructions to ensure optimal performance and longevity of the system.

1. System Components

- **Solar Panel:** Generates DC electricity from sunlight.
- **Bidirectional Inverter:** Converts DC to AC for loads and AC to DC for charging the battery.
- **Battery:** Stores energy for backup during outages.
- **Grid Connection:** Enables interaction with the main power grid.

2. Installation Steps

Tools Required:

- Screwdriver set
- Multimeter
- Wiring tools

Steps:

- **Solar Panel Setup:**
  - Mount the solar panel in a location with maximum sunlight exposure.
  - Connect the solar panel's output to the inverter's DC input.
- **Inverter Installation:**
  - Place the inverter in a well-ventilated area.
  - Connect the inverter to the solar panel, battery, and grid as per the wiring diagram.
  - Ensure proper grounding for safety.
- **Battery Connection:**
  - Connect the battery to the inverter’s designated terminals.
  - Ensure the battery polarity is correct.
- **Load Connection:**
  - Connect household appliances or other loads to the inverter's AC output terminals.
- **Grid Connection:**
  - Connect the inverter to the grid input using the appropriate wiring.

3. Operation Instructions

Normal Operation:

- During the day, the system will prioritize solar energy for powering loads and charging the battery.
- At night or during low sunlight, the system will use stored battery energy or draw power from the grid as needed.

Backup Mode (Grid Outage):

- The inverter will automatically switch to battery power during grid outages, ensuring uninterrupted supply to connected loads.

Energy Export:

- When the battery is full, and solar generation exceeds load demand, the system exports surplus energy to the grid.

4. Maintenance Guidelines

1. **Solar Panels:**
    - Clean the panels monthly to remove dust and debris.
    - Inspect for any cracks or damage.
2. **Inverter:**
    - Keep the inverter in a dry and ventilated location.
    - Check for loose connections and overheating.
3. **Battery:**
    - Replace the battery every 3–5 years or as recommended.
4. **Wiring:**
    - Inspect wiring connections periodically for wear and tear.
    - Replace damaged wires promptly.

5. Troubleshooting

Issue: No power output from the inverter.
Solution:

- Check solar panel and battery connections.
- Verify the inverter is properly connected to the grid.

Issue: Battery not charging.
Solution:

- Check solar panel output.
- Ensure the battery is not over-discharged or damaged.

Issue: Inverter overheating.
Solution:

- Ensure proper ventilation around the inverter.
- Reduce the connected load if it exceeds the inverter's capacity.

6. Safety Precautions

- Always disconnect the system from the grid and battery before performing maintenance.
- Do not expose the inverter to water or direct sunlight.
- Ensure proper grounding to avoid electrical hazards.

7.3 GitHub Link

https://github.com/DiptoArchyes07/Solar-bidirectional-inverter

7.4 YouTube Link

https://youtu.be/1hczdKhWUOE

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

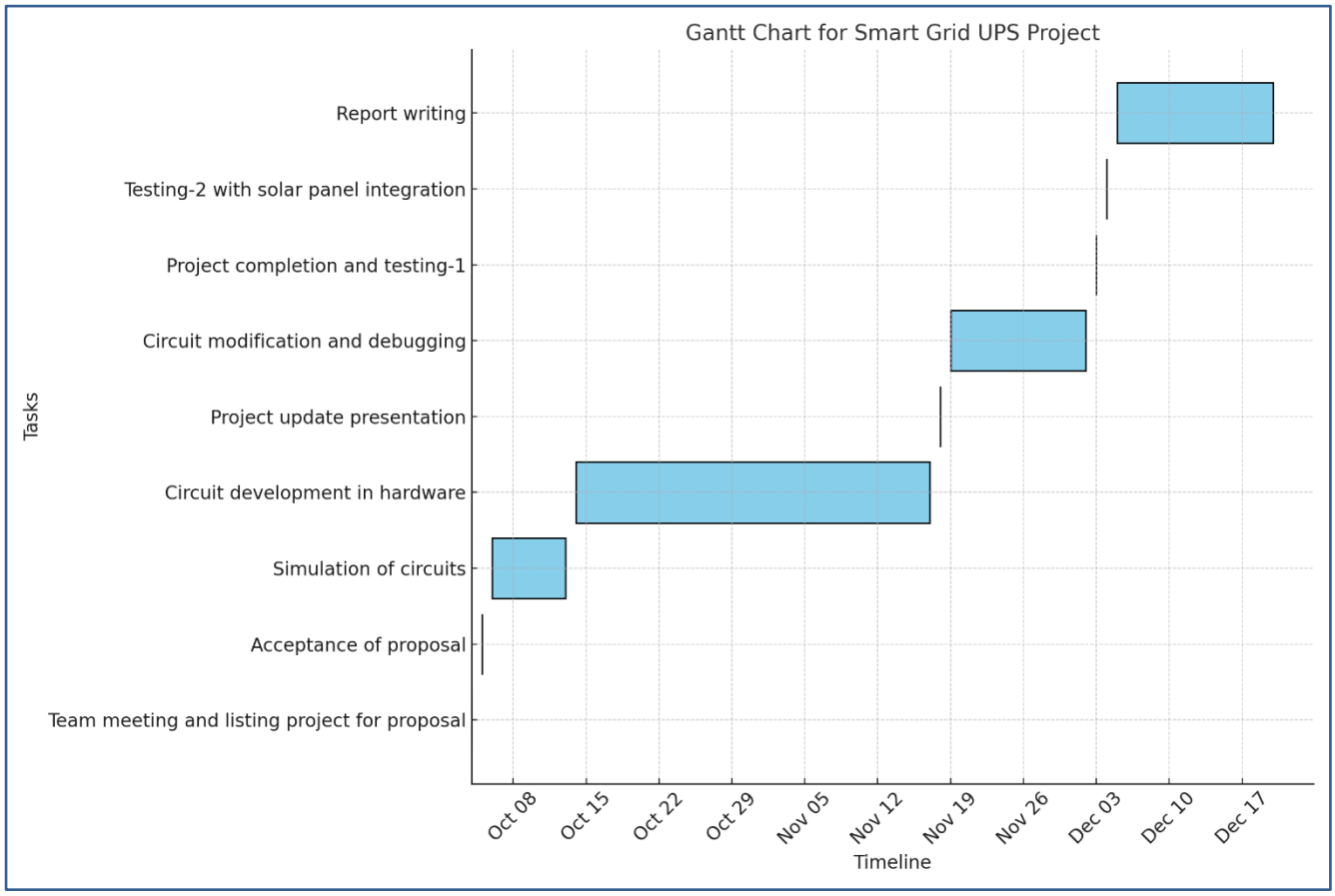
BOM of inverter components

Item	Quantity	Cost per Unit (PU)	Total Cost(Tk)
Breadboard	2	130	260
Center-tapped transformer	1	300	300
MOSFET IRFP250N	2	70	140
CD4047BCN IC - Pulse Generator	1	30	30
Switch SPST	1	10	10
Resistor (30 ohm, 100 ohm)	4	3	12
Capacitor (0.1uF, 47uF)	2	2	4
Diode 1N4001	2	2	4
Potentiometer - 100k	1	10	10
Wires (Jumper Set, Connector)	2	60	120
Connector Box	1	60	60
3W Bulb	1	180	180
Total			**1130**

Solar Charger Components

Item	Quantity	Cost per Unit (PU)	Total Cost (Tk)
Solar Panel	1	1800	1800
Breadboard	2	130	130
3.7V Rechargeable Battery	1	300	195
Battery Case	2	70	30
BJT 2N2222A	1	30	10
12V Relay	1	10	60
Resistor (220 ohm, 1k ohm)	4	3	6
Potentiometer - 5k	2	2	10
Diode 1N4001	2	2	4
Capacitor- 2200uF	1	10	20
LED- 2/3 color	2	60	120
Soldering Equipment (Iron, Lead)	1	60	200
Total			**2585**

8.2 Timeline of Project Implementation



9 Future Work (PO(l))

Different scopes are there for this project which can be progressed in future:

Sine Wave Inverter Implementation

The current project utilizes a basic bidirectional inverter to convert DC to AC. Future advancements could include the development of a sine wave inverter, which produces a smoother and more stable AC output. This would make the system suitable for sensitive electronic devices such as medical equipment, communication systems, and high-precision industrial tools. Sine wave inverters also improve energy efficiency and reduce harmonic distortion, resulting in better overall performance and longer lifespan for connected appliances.

Grid-Connected Applications

Expanding the project to integrate seamless grid-connected applications would significantly

enhance its utility. This involves creating a smart grid-interactive system capable of exporting surplus energy to the grid during low demand and importing energy during peak requirements. Such a system could contribute to grid stabilization and support renewable energy initiatives by increasing the share of solar power in the energy mix. Additionally, it could include smart metering and monitoring to track energy consumption and earnings from energy export.

Advanced Solar Tracking Techniques

Future iterations of this project could incorporate advanced solar tracking technologies to optimize energy generation. A dual-axis solar tracking system, for instance, allows the solar panels to follow the sun's movement throughout the day and across seasons, significantly increasing energy output. Implementing AI-based predictive algorithms could further enhance tracking accuracy, accounting for weather conditions and energy demand patterns. Such advancements would maximize the efficiency of the solar panels, ensuring optimal utilization of renewable resources and increased cost-effectiveness.

10 References

- Green, M. A., Emery, K., Hishikawa, Y., Warta, W., & Zou, J. (2010). Solar energy: The physics and engineering of photovoltaic conversion, technologies and systems. Wiley.

- Khan, I. M., & Ahmed, F. (2020). Design and performance analysis of a low-cost inverter for solar applications. Journal of Solar Energy Engineering, 142(5), 1-10.